IS 296 Course Project - Spring 2025

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Overview

In this course project, you will have the opportunity to design and implement a data-centric project to demonstrate your skills in understanding, processing, analyzing, and communicating with data.

What is Data Storytelling?

Data storytelling is to construct and present data stories to highlight the analytical insights in order to achieve the communication goals to a specific audience. It includes elements such as data visualization (e.g. in the form of data plots or animation) in conjunction with the narrative presented in formats such as presentation, video or creative formats such as data comics.

Project Teaming & Project Topic Selection

The project team can have 1-3 members. Students may form teams organically or with assistance from instruction teams based on common interests and backgrounds.

Timeline and Deliverables

Week Number	Tasks
Week 9	Finish teaming. Topic/data selection
Week 10 - 15	weekly mini update
Week 17	Final deliverable

Weekly mini-update (team-based)

- is due every Sunday at 11:59 pm, team-based product; however, each team member submits on the blackboard, simply type “done” if your team has finished mini-update by the deadline;
- Folder of mini-update, each team, please create a sub-folder with your “team ID_keyword”, for example, team S1 who work on healthcare should be named as “S1_healthcare”
- In this folder, create a google slide deck documenting the weekly progress of your team, mostly in the form of data insights (plot or table) you gathered from the Python Notebook and annotated with your analysis or interpretation (Show and Tell). Each team should aim for at least 2-3 data insights each week; at a later stage of the project, you will select and build a data story using a subset of those data insights. The instructor team (instructor, TFs) will give feedback to the team weekly. So please make sure that your slide decks are editable and commentable.

Diary Entries (individual)

- Each student will submit diary entries for each day that this student work on the project
- The diary entry ask students to document
    - Specific tasks worked on, time spent
    - Major accomplishments
    - Challenges encountered and resolution
    - Moments of discovery (‘aha’ moment)
    - Sentiment (select from drop down)

Biweekly short survey on motivation

Final-Deliverable

(a) Jupyter Notebook documenting data analysis process
(b) A slide deck containing a logical sequence of data insights with narratives
(c) A presentation recording in the form of video/animation that can be shared (optional)

Project Grading Criteria

- The identification of communication goals and audience is appropriate
- Each individual data insights
    - The choice of plot type is appropriate
    - The configuration of the plot is appropriate (e.g. data series, axis)
    - Narrative to explain the plot is correct and appropriate
    - If applicable, the “blank insight” is appropriate
- Storyboarding
    - The ordering and transition is appropriate
    - Overall the story is coherent and compelling
- Bonus
    - Use creative format such as data comics or data videos

Peer Review

For students who work in a team; each team member will submit a peer review form

Project Mentoring

Each project team will be assigned to one of the TAs/Data Scholars as a mentor. Mentors will provide guidance and technical assistance on data analysis tasks related to the project. The mentor and instructor could also provide feedback on the weekly mini-update. It is expected that each team should schedule at least one meeting with their mentor throughout the project.

Link to Pre-defined Project List

Project List

Appendix

Examples of Data Stories Videos

Explorer
https://videoexplorer.idvxlab.com/#
(all videos collected in a spreadsheet)

Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BBC Four

Examining Inequality by The Bill and Melinda Gates Foundation

30 Years of American Anxieties by The Puddling

NASA Goddard - What Are The Chances Of Another Katrina?

Profiling the Park

A collection of data stories

https://pyramidstory.idvxlab.com/#/Gallery

Example of Data Stories Videos by IS 296 students

Example digital data stories produced by students

The Story of World Happiness
Link to video https://youtu.be/-4XQqu-BXsU

In this story, the student walked through his analysis step by step toward the revelation of the major contributing factors to the country-level happiness score. His story unfolds by introducing the candidate factors one by one. Here, his use of stock images and videos appears quite effective. He then attracts the audience's attention by posing an intriguing question: Why some countries like Finland and Denmark scored the highest while countries like Afghanistan slipped to the bottom? Finally, he shared his aha moments of discovery. As he reflected at the end of the video on the meaning of happiness and potential pathways to improve world happiness: *"Happiness isn't just a commodity, it's a goal, something to strive for … from my findings, the proper steps needed will involve looking at the happiness driving factors for countries, like GDP, Social Support, Healthy Life Expectancy, and seeing to it that countries found on the lower end of the ladder positively achieve these. I feel that with this data, we can conclude, such as I have, that will point us to the kind of solutions and decisions needed to achieve true world happiness"*

The Story of Industrialization, Carbon Emissions, and Climate Change

https://youtu.be/g8qd-iQljBY

This story starts with a definition of carbon emission and plausible causes from industrialization, population growth, and resource overconsumption, which were later substantiated through data. The story presents insights by contrasting the amount of carbon emission from one developed and one underdeveloped country from each continent and follows with a historical account of industrialization and its impact on carbon emissions. The story further explores data showing the disproportional burden of air pollution-related health issues from developing countries as compared to developed countries. The story concludes with a statement of the consequences of

global warming, accompanied by a call to action by making small individual changes: "*Global warming threatens the survival of all living things and is directly linked to the health of the human species, especially in densely populated areas. Doing things like using reusable bags, using public transport instead of individual vehicles, and turning off electronic devices while not in use are some examples of small changes that many people can accomplish. If we all make small individual changes, we can accomplish great things together to ensure the health of future generations and our one and only planet.*"